

Article

A Hybrid Machine Learning Approach for High-Accuracy Energy Consumption Prediction Using Indoor Environmental Quality Sensors

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Abstract

Accurate forecasting of energy consumption in buildings is essential for achieving energy efficiency and reducing carbon emissions. However, many existing models rely on limited input variables and overlook the complex influence of indoor environmental quality (IEQ). In this study, we assess the performance of hybrid machine learning ensembles for predicting hourly energy demand in a smart office environment using high-frequency IEQ sensor data. Environmental variables including carbon dioxide concentration (CO₂), particulate matter (PM_{2.5}), total volatile organic compounds (TVOCs), noise levels, humidity, and temperature were recorded over a four-month period. We evaluated two ensemble configurations combining support vector regression (SVR) with either Random Forest or LightGBM as base learners and Ridge regression as a meta-learner, alongside single-model baselines such as SVR and artificial neural networks (ANN). The SVR combined with Random Forest and Ridge regression demonstrated the highest predictive performance, achieving a mean absolute error (MAE) of 1.20, a mean absolute percentage error (MAPE) of 8.92%, and a coefficient of determination (R²) of 0.82. Feature importance analysis using SHAP values, together with non-parametric statistical testing, identified TVOCs, humidity, and PM_{2.5} as the most influential predictors of energy use. These findings highlight the value of integrating high-resolution IEQ data into predictive frameworks and demonstrate that such data can significantly improve forecasting accuracy. This effect is attributed to the direct link between these IEQ variables and the activation of energy-intensive systems; fluctuations in humidity drive HVAC energy use for dehumidification, while elevated pollutant levels (TVOCs, PM_{2.5}) trigger increased ventilation to maintain indoor air quality, thus raising the total energy load.

Keywords: energy consumption; building energy management; machine learning; predictive modeling; indoor environmental quality (IEQ); support vector regression (SVR); hybrid models; smart buildings



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1. Introduction

The construction sector is the largest consumer of energy resources worldwide, accounting for roughly 36% of global energy use and about 39% of annual CO₂ emissions.

These figures place buildings at the heart of the global climate agenda and necessitate actions that cut energy demand across the building life cycle, accelerate renewable integration, deploy advanced insulating materials, and implement strict performance standards and smart controls to curb losses and operating costs [1]. Although a substantial share of these emissions is generated during the operational phase, a “hidden” carbon component is also associated with raw material extraction, material production, construction, and eventual demolition. In modern buildings, operational emissions exceed 70% of the total carbon footprint; however, as operational energy consumption declines, the share of embodied carbon becomes critically important [2]. Highly energy-intensive materials—reinforced concrete, steel, and brick—can contribute 60–70% of total emissions, underscoring the need for low carbon structural solutions and sustainable construction methods [1].

Carbon dioxide (CO₂) is one of the main greenhouse gases emitted as a result of anthropogenic activities such as fossil fuel combustion, cement production, and deforestation. It is a well-mixed gas in the Earth’s atmosphere with a complex removal mechanism involving physical processes like ocean absorption and chemical processes that occur over various timescales ranging from years to millennia [3,4].

Unlike reporting on net CO₂ emissions, carbon dioxide equivalent (CO₂-eq) is a metric developed to account for the overall warming effect of a mix of greenhouse gases. Not all greenhouse gases behave the same: gases such as methane (CH₄), nitrous oxide (N₂O), and various fluorinated gases exhibit different radiative efficiencies and atmospheric lifetimes compared to CO₂ [5].

The CO₂-eq approach involves converting emissions of non-CO₂ gases into an equivalent mass of CO₂ using global warming potential (GWP) factors. GWP is defined as the time-integrated radiative forcing of a pulse emission of a given gas relative to that of an equivalent mass of CO₂, typically aggregated over a 100-year time horizon (GWP100) [5].

A regional perspective reveals marked heterogeneity: in China alone, buildings generate nearly 5% of global CO₂ energy-related emissions, and similar proportions are evident in several developed countries where electricity supply still relies heavily on high-carbon-intensity fossil fuels [6]. Consequently, modernizing the existing building stock is among the most effective pathways to reduce energy consumption. For example, lowering air infiltration from 1.76 to 0.6 ACH in an office building saved 11% in energy costs and prevented roughly 20 t CO₂-eq annually; comprehensive thermal insulation of the envelope can raise total savings to almost 45%, equivalent to 75 t CO₂-eq per year [7].

Actual efficiency depends heavily on occupant behavior: deploying automated monitoring, smart controls, and energy use feedback systems delivers an additional demand reduction of up to 20% compared with design values [8]. Nonetheless, the performance gap between design and reality remains significant owing to technical constraints in the existing stock, high retrofit costs, and operational factors [8,9]. Financial barriers (capital scarcity, long payback periods, lack of incentives) and institutional fragmentation hinder large-scale adoption of best practices, while the absence of unified metrics complicates progress monitoring [9,10].

Energy efficiency forecasting is critical during design, operation, and retrofitting stages, yet the accuracy of most existing models is limited by their reliance on a small set of input variables such as outdoor temperature and time of day [11]. This simplification overlooks many internal processes and introduces systematic errors that render operational decisions sub optimal. Indoor climate emerges from nonlinear interactions among building materials, structural features, and occupant behavior [12]. These interactions include not only planned presence but also adaptive actions such as window opening, blind adjustment, and thermostat set point changes [13]. Behavioral responses are non-deterministic and vary both between different occupants and for the same individual over time, sharply reducing

the applicability of uniform behavioral assumptions [14]. Outdoor temperature and time of day capture basic trends in heating and cooling loads but cannot adequately represent internal fluctuations governed by occupants. Static models based on fixed timestamps or piecewise temperature regression ignore fast feedback loops between the microclimate and occupant actions, especially during abrupt condition changes [15]. Key parameters such as humidity, CO₂ concentration, and local thermal gradients are seldom incorporated [16]. High variability in occupant behavior intensifies discrepancies between models and actual consumption. Transient processes triggered by spontaneous actions generate rapid swings in thermal and internal quality loads that static timestamps fail to capture [17].

Dependence on a limited set of inputs such as outdoor temperature and time of day inevitably underestimates dynamic interactions between the indoor environment and occupant behavior. These overly simplified models cannot reflect the stochastic nature of behavioral responses, which include both adaptive and non-adaptive actions that substantially influence energy loads and indoor comfort [17]. Resulting performance gaps not only reduce the accuracy of energy consumption forecasts but also hamper optimization of building system designs and operational strategies [15].

Despite notable advances in hybrid machine learning models for energy consumption forecasting, a critical gap persists in developing and systematically evaluating multi-level hybrid architectures capable of simultaneously addressing both short- and long-term prediction horizons. Most studies focus on single models or shallow ensemble approaches without deep multi-level integration [18,19]. This deficit stems from the difficulty of combining multiple processing levels—each with distinct learning objectives and data requirements—into a unified composite structure. Convolutional neural networks effectively extract spatial features from structured sensor data or images, whereas recurrent architectures (LSTM, GRU) better capture temporal dependencies; however, coherent integration of these paradigms into a multi-level system remains a significant scientific and practical challenge [19,20].

Total volatile organic compounds (TVOC) are defined as the total concentration of volatile organic compounds (VOCs) in indoor air and serve as an aggregated measure without identifying individual chemicals. TVOC comprise a broad range of organic compounds that volatilize at room temperature, typically from C6 to C16 [21].

The main indoor sources of TVOC include building materials and finishes, adhesives, cleaning agents, paints, and various consumer products. In addition, outdoor contributions from industrial activity and vehicle emissions can increase indoor TVOC when they infiltrate through heating, ventilation, and air conditioning (HVAC) systems [22,23]. Secondary emissions also play a role; for example, spent fibrous filters in portable air cleaners can begin to release VOCs over time as equilibrium with the surrounding environment is reached [24]. Indoor TVOC concentrations vary with occupant activities, ventilation intensity, and environmental conditions such as temperature and humidity—factors that are critical for HVAC design and control [25,26].

Maintaining acceptable TVOC levels is a key objective for HVAC, as elevated concentrations are associated with adverse health effects, including airway irritation, headaches, and symptoms of sick building syndrome (SBS) [21,23]. While measurements of individual VOCs provide detailed chemical profiles, the TVOC metric offers a simplified, aggregate assessment of the overall chemical burden indoors and is convenient for rapid appraisal [22,23].

Heating, Ventilation, and Air Conditioning (HVAC) systems are designed to control indoor pollutant levels through appropriate ventilation and air filtration and, where needed, dynamic (demand-controlled) ventilation that modulates airflow in response to pollutant

concentrations. TVOC measurements provide direct feedback on the effectiveness of these ventilation strategies [27,28].

The eTVOC (equivalent TVOC) parameter is introduced to standardize and normalize the output of commercial sensors, which—being miniature devices designed for continuous monitoring—often cannot accurately separate or identify individual VOCs [29,30]. eTVOC is a calculated value obtained through an internal sensor algorithm that converts raw signals (changes in the conductivity of a metal-oxide element or currents generated during photo-ionization) into a number expressed in units equivalent to the concentration of a reference gas, typically toluene [5,31].

The fundamental difference between TVOC and eTVOC lies in how the data are obtained and interpreted. TVOC is the directly measured total concentration of all volatile organic compounds present in the air, determined with precise analytical methods such as GC/MS; it reflects the actual pollutant load, as each component is quantified and their concentrations summed using individual response factors [31,32]. In contrast, eTVOC is an output parameter provided by commercial sensors via internal signal-processing algorithms. These algorithms normalize raw sensor data, converting them into units equivalent to the concentration of the selected reference compound (toluene), thereby enabling comparability across devices despite differences in design, calibration mixtures, or operating principles [31].

Existing approaches to building energy consumption forecasting primarily rely on climatic, operational, and coarse occupancy parameters. In doing so, they overlook IEQ factors, such as noise level, PM_{2.5} concentration, and TVOC. Although these variables have been shown to influence occupant comfort and behavior, and consequently energy demand, they see limited use as direct predictors [31,32]. Even advanced hybrid schemes that blend observation and simulation data have not yet effectively integrated these multimodal IEQ signals. As a result, the development and objective evaluation of multilayer hybrid architectures capable of effectively utilizing data on noise, PM_{2.5}, and TVOC for energy forecasting remains an underexplored area [32,33], a gap that our research addresses.

2. Literature Review

Traditional building-energy forecasting methods rely on physical “white box” models that compute thermal and mass balances, sequentially describing all internal processes under specified conditions [34]. While this detail enables precise tracking of energy transfer mechanisms, such models poorly capture occupant behavior, which leads to significant errors and limits their usefulness for accurate load prediction [35]. A comprehensive physical model also demands a large volume of input data, and if these data are incomplete or inaccurate, the calculated results diverge from reality [36]. Educational buildings account for 10.79% of the total energy consumption of commercial buildings. Universities and colleges consume 33 billion kWh, representing 24.44% of the total energy use of educational institutions [37].

Traditional white box heat and mass balance models have gradually been strengthened with statistical regression, giving rise to black box solutions that retain acceptable accuracy while demanding only modest computational resources. A vivid example is the three-parameter heating and cooling degree model, whose key coefficients are derived from historical weather data and energy consumption records [38,39].

The earliest machine learning studies in energy consumption forecasting applied SVMs and multilayer perceptions, training them on load, weather, and calendar features. SVMs outperformed classical regression in cooling load prediction thanks to their ability to handle high-dimensional data and built-in regularization [40]. Subsequent advances introduced deep networks and LSTMs that capture temporal dynamics, markedly reducing mean

absolute errors compared with traditional methods [29,41]. ML approaches offer flexibility, require minimal detailed engineering input, and can self-learn as data accumulate, which is vital amid stochastic demand fluctuations and changing occupant behavior [42].

The widespread deployment of IoT sensors now provides continuous, high-frequency measurements of temperature, humidity, occupancy, illuminance, and equipment status. These streams feed BMS and BACS platforms, supporting calibration of physics-based models and training of machine learning algorithms [43,44]. Cloud computing and standardized protocols such as BACnet have expanded analytical capabilities, enabling predictive maintenance and real-time adaptive control [45]. External meteorological variable solar radiation, wind, and air temperature are fused with indoor measurements, boosting forecast accuracy and enabling proactive responses to extreme weather swings. Sensor data has thus become a critical foundation for flexible energy strategies, underpinning both physics-based and data-driven approaches [45,46].

When data are limited or noisy, individual ML models tend to overfit, so hybrid and ensemble solutions are gaining traction. Aggregating forecasts from ANNs, SVMs, and rule-based models such as M5Rules reduces bias and variance, lowering MAE and MAPE relative to single algorithms [30,47]. In particular, it achieved the lowest MAPE (3.264) and the highest R^2 score (0.966), demonstrating its effectiveness [30]. Ensembles can dynamically re-weight components during abrupt shifts in occupant behavior or weather, delivering more than a two-fold accuracy gain in some studies [47,48]. Applying transfer learning can reduce 24 h building energy consumption forecasting error by approximately 15–18% [49].

The synergy of traditional physical modeling, machine learning, sensor technologies, and ensemble strategies forms the backbone of next generation energy consumption forecasting systems. Continued interdisciplinary research and collaboration among academia, industry, and regulators will foster cost-effective, environmentally sustainable solutions that set new benchmarks for energy conservation in the built environment [32,50,51].

Data-driven methods have superseded purely physical approaches by exploiting historical measurements to identify key factors in energy use through machine learning and deep learning algorithms [52]. ANNs, especially multilayer perceptions, approximate nonlinear relationships well but require substantial computing resources and provide low interpretability [53]. SVMs deliver stable results with any data volume yet incur high resource demands and limited computational efficiency, and, in several studies, they outperform neural networks [54]. The Random Forest algorithm has proven to be a precise and reliable tool, particularly for long-term forecasts of electricity and heat consumption [55]. Long short-term memory networks capture long-range dependencies in time series and surpass classical statistical models such as ARIMA and SVR in seasonally sensitive predictions. Convolutional neural networks, although relatively new to this domain, effectively extract spatiotemporal patterns and offer high computational efficiency [53]. XGBoost has emerged as one of the best methods for forecasting energy demand and prices, outperforming many other models in both accuracy and speed [56]. ARIMA models are applied to short-term forecasts but lag behind LSTM on longer horizons, whereas simple linear regression remains optimal for niche tasks such as annual forecasts [53,57]. Ultimately, no algorithm is universally superior, and the choice depends on task conditions and data characteristics [54].

The rise of IoT and smart buildings has generated continuous streams of sensor data that are critical for improving forecast accuracy [30,58]. Inputs now include historical consumption records, calendar features, weather conditions, occupancy and behavioral information, building geometry and construction properties, equipment and system loads, and socioeconomic factors [59]. These data arrive from building management systems,

smart meters, and weather stations, yet often contain noise, missing values, and heterogeneous formats, so data cleaning, feature extraction, and integration techniques remain essential for model reliability [53,60]. Recursive feature elimination (RFE) helps analyze feature importance for energy consumption prediction; it was found that the top 10 features account for over 80% of the total importance [61].

Hybrid models in energy forecasting merge the complementary strengths of diverse modeling paradigms to capture both linear and nonlinear characteristics of energy data [62]. Studies show that integrating statistical time series techniques—such as SARIMA—with machine learning algorithms yields more accurate forecasts by first modeling residual errors or decomposing the series into distinct frequency components before applying nonlinear learners [63]. A frequently cited example is the SARIMA-MetaFA-LSSVR hybrid, which surpasses single models by delivering higher correlation coefficients and lower mean absolute percentage errors when predicting residential-building energy consumption [64].

Ensemble methods have been successfully employed in energy forecasting to predict diverse outcomes such as global solar radiation, residential energy consumption, and wind-power generation. A multi-level ensemble that combines support vector regression, artificial neural networks, and linear regression yields robust short-term load forecasts by compensating for the individual weaknesses of each component model. In addition, ensembles have proven effective with highly variable datasets by integrating predictions across multiple forecast horizons, resulting in lower mean absolute percentage error (MAPE) and coefficient of variation values [63]. The hybrid CNN–LSTM model achieved a MAPE of 4.58% on clear days and 7.06% on cloudy days for forecasting PV power output [65].

A growing research focus is the development of advanced control strategies that use computational intelligence to optimize energy efficiency while maintaining high indoor environmental quality (IEQ). Many researchers have adopted methods like Model Predictive Control (MPC), reinforcement learning, and multi-agent systems. Several studies propose Model Predictive Control (MPC) frameworks that incorporate weather forecasts, dynamic pricing signals, and occupancy profiles to adapt HVAC operations in real time [66]. These methods not only aim to reduce energy usage by shifting or shedding HVAC loads but also integrate thermal comfort metrics—direct measures of IEQ—into the control objectives. This confirms the idea that successful energy management strategies in grid-connected microgrid environments must be both demand-driven and occupant-centered [66].

At the system level, various algorithms have been proposed to manage these competing objectives across multiple buildings. The study by Baldi et al. proposes a novel control algorithm that jointly optimizes energy demand and thermal comfort in interconnected microgrids equipped with photovoltaic (PV) panels [67]. By incorporating real-time occupancy data and leveraging the PCAO (Parameterized Cognitive Adaptive Optimization) algorithm, the proposed EMS achieved up to 36% total cost improvement compared to rule-based strategies and maintained thermal dissatisfaction (PPD) levels below 7% across different buildings and weather conditions [68]. This approach introduces a simulation-based optimization framework relying on detailed building energy models developed in EnergyPlus 25.1.0. The algorithm adjusts the HVAC setpoints across multiple buildings to balance the availability of solar energy and minimize grid absorption while maintaining thermal comfort, evaluated using indices such as Fanger’s model or the Predicted Mean Vote (PMV) [67]. In a similar vein, Korkas et al. presented a demand-side management strategy that optimizes thermal comfort in microgrids integrating renewable energy sources and energy storage systems [69]. Their research introduces a scalable two-level dispatch control scheme: local controllers autonomously regulate HVAC systems based on real-time occupancy and temperature data, while a higher-level controller oversees the global energy management. This hierarchical structure effectively balances the competing

objectives of minimizing electricity costs and ensuring occupant comfort, particularly by accommodating spatial variability in occupancy patterns across different building zones.

These optimization principles are also being applied in the residential sector. Anvari-Moghaddam et al. developed an efficient energy management system (EMS) for residential grid-connected microgrids, which integrates distributed generation models, building thermal dynamics, and load-scheduling capabilities for household appliances [70]. While the primary focus is on reducing household energy consumption and electricity costs by scheduling loads according to real-time constraints and user-defined deadlines, the EMS also incorporates user comfort measures. Thus, this approach implicitly connects energy efficiency efforts with indoor thermal comfort—one of the most critical aspects of IEQ—by ensuring that scheduling practices consider not only energy optimization goals but also the thermal and, indirectly, visual and acoustic comfort needs of occupants.

Ultimately, the effectiveness of these systems depends on understanding and predicting human behavior. Occupant behavior modeling is a subject of intense study, with researchers like Langevin et al. proposing the Human and Building Interaction Toolkit (HABIT), an agent-based model integrated with platforms like EnergyPlus [71]. Based on perceptual control theory, the HABIT model simulates how occupants adjust their behavior in response to thermal sensations. The results show that active, adaptive occupant behavior significantly impacts both energy consumption and the resulting IEQ, highlighting the importance of incorporating behavioral dynamics into smart energy management system strategies [71].

The study by Natarajan et al. introduces a hybrid CNN_BiLSTM model that combines convolutional filters with bidirectional LSTM cells and is trained on high-granularity IoT smart-meter data to forecast weekly energy consumption in residential and commercial buildings. In its optimal configuration, the model attains $MSE = 4570.14$, $MAPE = 4.98\%$, $RMSE = 67.60$, $MAE = 46.10$, and $R^2 = 0.96$, demonstrating high accuracy for long-term prediction. Compared with standalone CNN, BiLSTM, CatBoost, LGBM, and XGBoost models, its mean percentage error is reduced by roughly a factor of five, and its superiority across all metrics persists throughout the entire forecasting horizon, confirming the robustness of the hybrid approach. These results strengthen the theoretical rationale for employing hybrid architectures and conducting feature-importance analysis in the present manuscript [72]. A comparative evaluation of XGBoost, SVR, and LSTM for multi-year building energy forecasting in a hot climate showed that LSTM delivers outstanding accuracy ($R^2 = 0.993$; $RMSE = 0.004$), markedly outperforming SVR ($R^2 = 0.462$) and XGBoost ($R^2 = 0.94$); this high effectiveness is attributed to the recurrent network's ability to capture prolonged nonlinear dependencies among construction characteristics, meteorological factors, and operational regimes [73]. A genetically optimized fuzzy regression model is also introduced, generating interpretable rules and achieving the lowest prediction errors for residential electricity consumption, outperforming 20 alternative algorithms, including gradient boosting and deep neural networks [74].

A comparative analysis of 23 machine learning algorithms applied to daily wastewater-treatment-plant data showed that shifting from “static” schemes to dynamic ones—including a one-day load lag among the inputs—significantly boosts forecasting accuracy; after hyperparameter tuning, this setup pushed k-nearest neighbors to the top with $RMSE \approx 37$ MWh and $MAPE \approx 10.7\%$, whereas the heavier XGBoost brought only a marginal error gain at a 30-fold increase in training time [75]. On a dataset from five university buildings, a different ensemble—a voting scheme combining ANN, SVR, and M5Rules—delivered average $MAE = 2.86$ kWh, $MAPE \approx 16\%$, and $RMSE \approx 4$ kWh, more than halving absolute error and cutting percentage error threefold versus the best single models [76]. A short-term Random Forest model trained on five year-long hourly datasets achieved $MAE = 0.430$ – 0.501 kWh

for the 1-step horizon, 0.612–0.940 kWh for 12 steps, and 0.626–0.868 kWh for 24 steps; its accuracy surpassed both Random Tree and M5P, improving MAE and MAPE by roughly 49% at one step ahead and 29–50% at longer horizons, confirming the suitability of tree ensembles for real-time load management across multiple buildings [77]. Benchmarking artificial neural networks, gradient boosting, deep neural networks, Random Forests, stacked ensembles, k-nearest neighbors, support-vector machines, decision trees, and linear regression on a large corpus of residential-design data revealed the clear supremacy of the deep neural network: the lowest MAE and MAPE together with the highest R^2 endorse its use for precise annual energy-use prediction. The model ingests key conceptual parameters—floor area, orientation, envelope characteristics, climate—and instantly estimates expected consumption at the early stage, enabling architects to optimize designs and curb future energy costs before construction begins [78]. Streaming data from a smart building confirmed that coupling an ML forecast with an optimization layer minimizes error and is more practical than single or ensemble methods: the hybrid provides the highest accuracy and is best suited for real-time energy management and operating-cost reduction [79].

A review of existing research reveals a multitude of approaches to forecasting energy consumption in buildings, ranging from classical statistical models to advanced machine learning ensembles. To systematically organize these approaches and clearly define the position of our research, Table 1 presents a comparative qualitative analysis of seminal and recent works in this field. The table is structured to allow the reader to assess the model architectures used, the types of input data, and the primary scientific contributions of each study. A key element of the table is a column that contrasts the novelty of each approach with our own research. This comparative analysis not only summarizes the accumulated knowledge but also clearly demonstrates the relevance of our contribution, which lies in integrating high-frequency indoor environmental quality (IEQ) data with hybrid stacked ensembles to improve forecast accuracy.

In addition to the analysis of methodologies, Table 2 summarizes the quantitative performance metrics of the models as reported in the reviewed studies. This analysis makes it possible to evaluate the level of accuracy achieved in various application areas—from wastewater treatment plants to residential buildings. The table presents key error metrics such as RMSE, MAE, and MAPE.

Despite considerable progress in smart building analytics, a literature review highlights a gap in the limited use of IEQ data for energy prediction. This study aims to investigate an existing gap in smart building analytics related to the limited use of indoor environmental quality (IEQ) data for energy consumption forecasting. It considers the application of comprehensive IEQ data as a significant source of input data. The paper proposes an interpretable hybrid ensemble model that shows higher accuracy compared to standard algorithms when processing high-frequency IEQ signals, such as noise, $PM_{2.5}$, and volatile organic compounds (TVOC). The predictive role of these variables is assessed using SHAP analysis. The model's accuracy allows for its use in the adaptive control of HVAC (Heating, Ventilation, and Air Conditioning) systems to improve energy efficiency without compromising occupant comfort. This allows IEQ data to be viewed not only as a comfort metric but also as a forecasting tool.

Table 1. A comparative analysis of the used methods and the novelty of existing research.

Reference	Research/Model Type	Architecture/Model Used	Input Data Used	Main Contribution/Objective	Distinction and Novelty of Our Work	Comparison with Our Novelty
[75]	Comparative ML (static vs. dynamic) for Wastewater Treatment Plan (WWTP) energy prediction.	23 ML models including Random Forest, XGBoost, and dynamic KNN with time-lagged features; Bayesian optimization.	Real operational data from a WWTP (aeration, hydraulic, climate, wastewater quality).	To demonstrate that dynamic models with lagged features outperform static ones for improved energy management.	Integration of multifaceted sensor data and dynamic modeling to capture nonlinear, time-dependent WWTP behavior.	Our novelty is in the building IEQ domain and a specific stacked model, differing from their focus on industrial process control in a WWTP.
[79]	Comparative forecasting study of single, ensemble, and hybrid AI models for residential energy.	ANN, SVR, CART, SARIMA, and hybrid SARIMA-LSSVR models with metaheuristic optimization.	High-frequency residential energy time series with usage patterns and temporal indices.	To develop and assess models to provide accurate next-day forecasts for residential energy consumption.	A novel combination of SARIMA and metaheuristic-optimized LSSVR, capturing both linear seasonal and nonlinear patterns.	Our novelty lies in a different hybrid structure (stacking) and using rich IEQ inputs (TVOC, PM _{2.5}), not a SARIMA-based hybrid focused on residential usage patterns.
[76]	Ensemble ML approach for building energy consumption prediction.	Integration of ANN, SVR, and M5Rules via an averaging/voting ensemble method.	Hourly energy consumption data with temporal features from educational/non-residential buildings.	To enhance 24 h energy forecast accuracy by leveraging ensemble techniques over individual predictors.	Systematic integration and fine-tuning of heterogeneous models, achieving lower error metrics.	Our model is a more complex multi-level stacking ensemble, not a simple averaging/voting ensemble. Our novelty is also in the unique IEQ input data.
[78]	Comparative study employing deep learning and ML for annual energy prediction in residential buildings.	Multilayer perceptron ANN, SVM, and decision tree regression models.	Data from ~300 residential buildings, including building metadata and meteorological variables.	To evaluate deep learning (ANN) versus traditional models for early-stage design applications.	A focus on early-stage prediction using deep learning, offering an alternative to simulation tools.	Our novelty is in hourly operational forecasting using granular sensor data, not annual prediction for the design phase using high-level building metadata.
[77]	Multi-building energy consumption prediction framework for sustainability.	Primarily tree-based ensemble methods, particularly Random Forest (RF).	Hourly energy consumption data across multiple buildings with operating schedules and time indices.	To develop a scalable forecasting framework for energy efficiency across multiple buildings.	Leveraging ensemble tree models (RF) to accurately capture heterogeneous multi-building data and temporal patterns.	Our novelty is in the depth of analysis for a single, deeply instrumented space to prove the value of IEQ inputs, rather than focusing on scalability.
[63]	Comprehensive review of ML/DL models for forecasting renewable energy generation.	A survey of classical regression (SVM, RF) alongside deep learning architectures (CNN, RNN, LSTM).	Meteorological time series data and historical renewable energy outputs.	To synthesize state-of-the-art advancements and outline future research directions in renewable energy forecasting.	Distinguished by its broad and integrative review of both classical and deep learning methods.	Our work is an empirical study proposing a new predictive model for consumption, not a literature review on renewable energy generation.

Table 1. Cont.

Reference	Research/Model Type	Architecture/Model Used	Input Data Used	Main Contribution/Objective	Distinction and Novelty of Our Work	Comparison with Our Novelty
[66]	Mixed-method review of thermal comfort and occupant behavior impacts on building energy consumption.	A review of various modeling approaches: statistical, simulation-based, agent-based, and ML models.	Aggregated literature data including indoor comfort measurements, occupant surveys, and building energy records.	To provide an integrative synthesis of how thermal comfort and occupant behavior influence energy use.	Unique for combining quantitative scientometric analysis with qualitative review, bridging technical models and human factors.	Our study provides empirical validation for the concepts reviewed by using IEQ data (a proxy for comfort/behavior) to directly predict energy use.
[67]	Joint optimization study integrating energy demand management with thermal comfort in microgrids.	An integrated optimization framework (likely involving mathematical programming and/or metaheuristics).	Simulation or real data including PV generation profiles, demand curves, and thermal comfort indices.	To develop a model that simultaneously minimizes energy consumption while satisfying occupant thermal comfort.	A novel integration of renewable energy generation and occupant comfort metrics in the optimization of interconnected microgrids.	Our novelty is in accurate prediction/forecasting, not control or optimization. Our model could serve as an input to an optimization framework like theirs.

Table 2. Performance summary of ML models in the literature.

Application Domain	Model	Forecast Horizon/Resolution	RMSE	MAE	MAPE (%)	Reference
Wastewater Treatment	XGBoost (Dynamic)	Dynamic model	37.14 MWh	28.50 MWh	10.81	[75]
Residential Buildings	Hybrid SARIMA-LSSVR	Hourly predictions	0.164 kWh	0.028 kWh	15.66	[79]
Educational Buildings	Ensemble Model	Hourly predictions	4.006 kWh	2.858 kWh	16.14	[77]
Educational Buildings	SVR	Hourly predictions	4.167 kWh	3.034 kWh	17.02	[77]
Annual Consumption	DNN	Annual forecast	2.80	2.20 (abs)	-	[78]
Short-Term Forecast	Random Forest (RF)	1 step ahead	0.648 kWh	0.499 kWh	6.02	[77]

3. Materials and Methods

3.1. Data Acquisition

Measurements were conducted over a four-month period in a 48 m² smart laboratory at Al-Farabi Kazakh National University. The room emulates an open-plan office with eight workstations, ceiling-mounted LED luminaires, low-density plug loads, and a full suite of HVAC equipment—three inverter split-AC units, two ultrasonic humidifiers, and a 240 m³ h^{−1} recirculation fan. All end uses are switched through Aqara Smart Plugs so that the same Zigbee 3.0/Wi-Fi mesh that carries sensor traffic can also enforce set points issued by the Home Assistant gateway or by voice commands.

Environmental and electrical variables were logged continuously from 16 March to 16 June 2025. Raw acquisition comprised 2928 h (≈12 weeks) of second-by-second indoor-air-quality records and minute-resolution plug-level power data. Time-stamped Zigbee frames and HTTPS payloads were ingested by a Raspberry Pi 4 running Home Assistant, normalized to UTC, and written to a MariaDB instance; daily export jobs replicated cleansed tables to a PostgreSQL server. Device clocks synchronized with an on-premises NTP source every 10 min, keeping worst-case drift below 10 ms throughout the campaign. After deduplication, spike filtering, and gap-filling, 1602 complete hourly records remained for downstream modeling.

Table 3 summarizes the 10 sensing and metering devices deployed in the laboratory together with the variables, units, and sampling rates delivered by each instrument. Accuracy figures are taken from the manufacturers' data sheets and were verified during periodic gas-zero, salt-bath, and power-analyzer checks. The resulting multi-variable stream resolves thermal, gaseous, particulate, acoustic, occupancy, and energy phenomena at 1 Hz, which is sufficient to capture rapid IEQ excursions and HVAC responses. Each row corresponds to a physical device, while the columns convey the manufacturer and model (with quantities where applicable), the environmental or electrical variables measured along with their units, the manufacturer-reported accuracy confirmed by in-house verification, the sampling rate after timestamping, and notes on mounting location or monitored end use.

Table 3. Sensors and devices in the smart laboratory.

No.	Device	Variables and Units	Accuracy	Sampling Frequency	Notes
1	Xiaomi Qingping Air Monitor 2 (CGS2)	Temp (°C), RH (%), CO ₂ (ppm), PM _{2.5} /PM ₁₀ (µg m ^{−3}), TVOC (mg m ^{−3})	CO ₂ ± 50 ppm; Temp ± 0.3 °C; RH ± 3%	1 Hz	Two units, ceiling-mounted east and west
2	Xiaomi Qingping Air Monitor 2 Pro	Same as CGS2	Same as CGS2	1 Hz	Two units, north and south walls
3	Aqara Temp & Humidity Sensor T1	Temp (°C), RH (%), Pressure (kPa)	Temp ± 0.3 °C; RH ± 3%; P ± 0.12 kPa	1 Hz	Four units at desk height
4	Aqara Air Quality Monitor	TVOC (mg m ^{−3}), Temp (°C), RH (%)	TVOC ± 0.01 mg m ^{−3}	1 Hz	Central column
5	Winsen ZPHS01B	CO ₂ (ppm), PM _{2.5} (µg m ^{−3})	CO ₂ ± 50 ppm	1 Hz	Duct probe in recirculator
6	Aqara Smart Plug (×8)	Power (W), Energy (kWh) for ACs, fans, humidifiers	±2% of reading	60 s	ACs (3), recirculator, humidifiers (2), lighting bus, spare
7	Aqara Presence Sensor FP2 (×5)	Occupancy (binary + zone ID)	—	Event-driven (~1 Hz heartbeat)	Ceiling-mounted grid

Table 3. Cont.

No.	Device	Variables and Units	Accuracy	Sampling Frequency	Notes
8	Aqara Door/Window (×3)	Door state (open/closed)	—	Event-driven	Main door + 2 windows
9	MEMS microphone (in each Qingping)	A-weighted sound level (dB)	±1.5 dB	1 Hz	Acoustic comfort proxy
10	Voice-assistant mic (Home Assistant)	Wake-word and command log	—	Event-driven	Adds user-interaction context

Figure 1 summarizes the cyber-physical layout that underpins data acquisition and automated control in the university smart laboratory. All edge devices appear on the left-hand side of the diagram. Environmental conditions are captured by a heterogeneous cluster of instruments: a Winsen ZPHS01B module and an MI Qingping monitor (Beijing, China) provide measurements of CO₂ concentration, particulate matter, temperature, and relative humidity; an Aqara TVOC Air-Quality Monitor refines gaseous-pollutant readings; an Aqara Temperature-and-Humidity T1 sensor augments the thermal field with barometric pressure; and Aqara Motion, Presence FP2, Door/Window and Water-Leak sensors contribute occupancy, safety, and context signals. An additional Aqara Smart Plug is installed at the edge layer (left-center of the figure); although drawn among the sensors, it functions as a metering point for miscellaneous laboratory appliances and is therefore grouped with the sensing infrastructure.

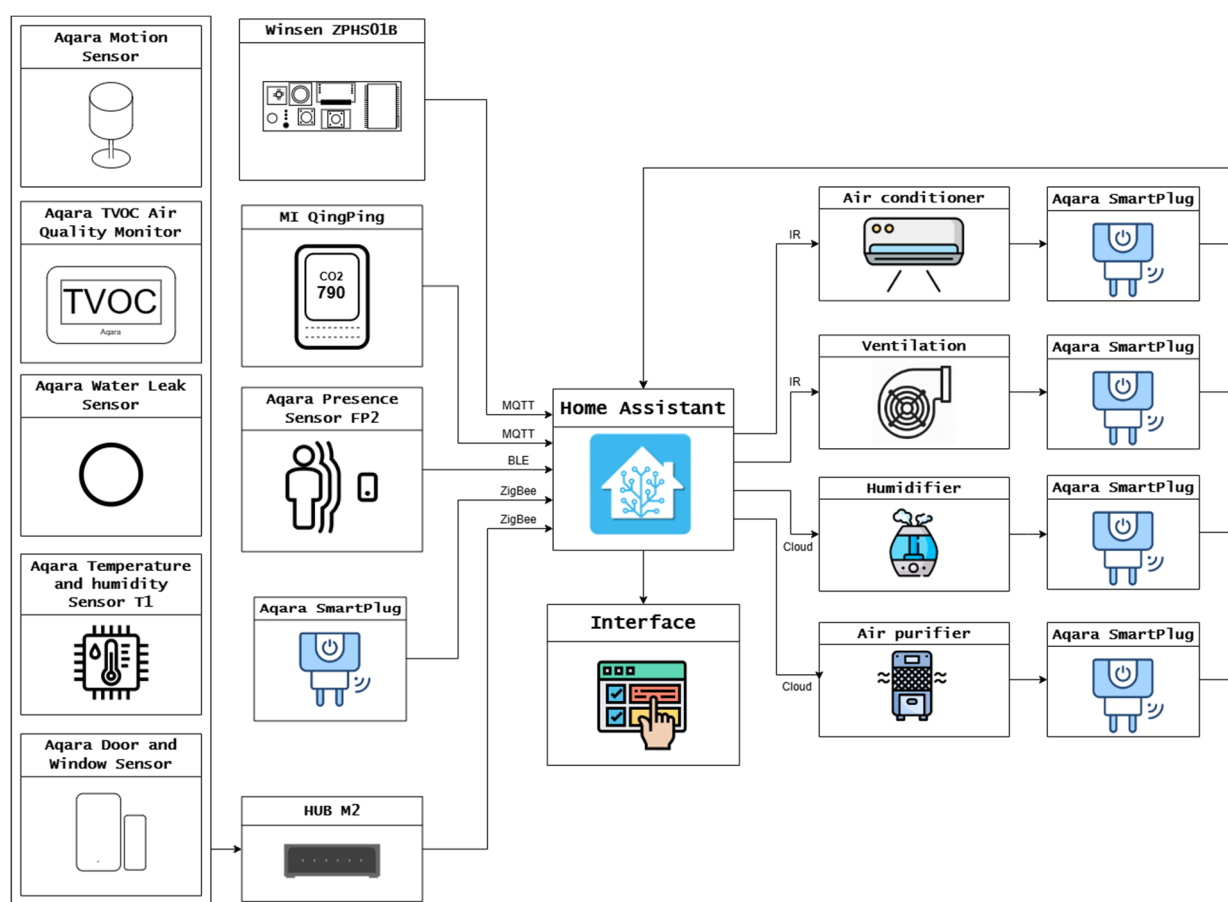


Figure 1. Smart laboratory IoT sensing-and-control architecture.

Connectivity paths converge on the central Home Assistant instance, shown in the middle of the scheme. A suite of Zigbee-based Aqara devices—specifically the Motion, TVOC Air Quality, Water Leak, Temperature/Humidity T1, and Door/Window sensors—are aggregated by an Aqara Hub M2, which in turn integrates with Home Assistant. The Winsen module and the MI Qingping monitor publish data via MQTT, while the Aqara Presence Sensor FP2 uses a combination of Wi-Fi and Bluetooth LE (BLE). Both the upstream (sensor-to-hub) and downstream (hub-to-actuator) links are explicitly labeled in the diagram with their transport layers—MQTT, BLE, Zigbee, IR, and Cloud—highlighting the heterogeneous yet interoperable nature of the network stack. The Aqara Smart Plug fleet—one in the sensing column and five on the actuator side—also forms part of the Zigbee mesh and is routed through the same Hub M2, reinforcing network resilience thanks to their repeater capability.

Home Assistant acts simultaneously as gateway, time series logger, and rule engine: it normalizes payloads received from all sources, stamps every record to UTC and exposes a REST/MQTT API for higher-level services. Internally, Home Assistant writes each measurement to an InfluxDB time series database and employs an event-driven automation engine that evaluates logical rules every second, ensuring sub-second response latency. All MQTT traffic is secured with TLS encryption to safeguard data integrity and confidentiality. From this hub, two control channels branch rightwards. Infrared (IR) commands drive a split-system air conditioner (“Air conditioner”) and the main ventilation fan (“Ventilation”), while HTTPS cloud calls adjust the ultrasonic humidifier (“Humidifier”) and a HEPA-equipped air purifier (“Air purifier”). Each end use is routed through an Aqara Smart Plug, enabling real-time measurement of instantaneous power and cumulative energy as well as on-demand load shedding. These smart plugs also communicate via Zigbee, connecting directly to Home Assistant’s coordinator.

A graphical interface, positioned beneath the Home Assistant block in the figure, delivers live dashboards, comfort warnings, and voice-assistant intents to users, closing the loop between sensing, analytics, and actuation. The interface component further supports a Prometheus endpoint for research-grade data export and a kiosk-mode dashboard on a wall-mounted tablet, mirroring the layout of Figure 1. Taken together, the layered architecture guarantees millisecond-accurate synchronization of environmental and energy data streams, while still supporting automated or voice-triggered control of all HVAC and ancillary systems—thereby providing the empirical backbone for the laboratory’s energy-efficiency and comfort-modeling research.

3.2. Data Preprocessing

Throughout a carefully monitored four-month campaign, the experimental laboratory logged high-resolution streams of carbon dioxide, relative humidity, fine particulate matter $PM_{2.5}$, total and equivalent volatile organic compounds, temperature, and cumulative energy. Alert-oriented channels such as instantaneous power, water leak, smoke, and dual motion sensors contained no usable scientific signal because more than 70% of their rows were empty and the binary alerts risked revealing the cumulative-energy target; therefore, these columns were eliminated at the outset. The retained eight variables captured the essential dynamics of indoor air quality and microclimate and provided a consistent physical basis for energy demand modeling.

All headers were lower-cased, whitespace-trimmed, comma decimals-converted, and values-cast to numeric. Data were resampled to the native interval, and rows with missing values in any of the eight features were dropped. Studies show that when over 50% of a channel is missing—often due to burst outages—list-wise deletion with robust scaling

outperforms imputation [80]. The timestamp string, which caused parsing errors, was removed after alignment.

A reproducible cleaning stage then excised stray punctuation, surplus whitespace, physically impossible magnitudes, and duplicate entries. Automated, code-controlled sanitation is strongly recommended for multichannel time series because it prevents silent spreadsheet edits and guarantees full auditability of every transformation [81].

Feature scaling employed the Z-score, defined as

$$z_i = \frac{x_i - \mu_{train}}{\sigma_{train}} \quad (1)$$

where x_i is the raw sensor value, while μ_{train} and σ_{train} are the mean and standard deviation fitted only on the training partition to avoid information leakage. The scaling parameters were computed from the training set and subsequently applied to the training, validation, and test data to avoid data leakage. Z-score normalisation brings heterogeneous sensors onto a common scale, accelerates gradient descent, and yields more balanced influence among variables inside kernel and ensemble learners, advantages that have been demonstrated on airborne pollutant, structural health, and biomedical telemetry [82].

The final design matrix therefore consists of seven fully standardized predictors and one standardized target, each observation precisely aligned in time and free from gaps, outliers and leakage. This high-quality foundation underpins the regression ensembles presented in the subsequent sections.

3.3. Machine Learning Models

To predict energy consumption under laboratory conditions, we explicitly modeled the energy sensor reading as the target variable and used seven environmental channels—CO₂, humidity, noise, PM_{2.5}, temperature, TVOC, and eTVOC—as predictors. Of the available data, 80% were used for training and the remaining 20% were held out for testing. Within the 80% training set, a robust validation strategy was employed for hyperparameter tuning and model selection. Specifically, a five-fold cross-validation technique was implemented using the GridSearchCV function. To eliminate split-selection bias and obtain reproducible generalization estimates, all models were trained using a stratified five-fold cross-validation (5-fold CV) protocol. Single learners, including SVR and the artificial neural network (ANN), were tuned via GridSearchCV, which internally applies 5-fold CV to select the hyperparameters that maximize mean validation R². For the ANN, the KerasRegressor was wrapped in GridSearchCV, and early stopping was applied within each fold to prevent overfitting. The selected architecture (two hidden layers with 64 and 32 neurons, ReLU activation, dropout 0.2, Adam optimizer with $\eta = 10^{-3}$) was chosen based on cross-validated performance. In the StackingRegressor, the same outer folds were reused to generate out-of-fold predictions from each base learner (SVR, Random Forest, LightGBM), which served as input to the Ridge meta-learner. Ridge regularization (α) was tuned via a nested 5-fold CV on these stacked predictions, ensuring strict separation between base and meta training. All folds were shuffled (shuffle = True) with a fixed random seed (42) to ensure reproducibility. This process automatically partitioned the training data into five equal folds, iteratively training the model on four folds while validating its performance on the remaining hold-out fold. This cycle was repeated five times, ensuring each fold served as the validation set once, which provided a stable evaluation to determine the optimal hyperparameters. The 20% test set was kept completely separate during this entire process and was used only at the very end to provide a final, unbiased assessment of the best model's generalization performance. Three modeling strategies were evaluated: support vector regression (SVR), a feed-forward artificial neural network (ANN), and two hybrid stacked-ensemble models.

These choices span kernel-based, neural, and stacked paradigms, giving us theory-driven baselines and a complementary hybrid design [62]. SVR serves as a baseline nonlinear regressor, capable of capturing complex and skewed dependencies in the sensor data [83]. It solves an optimization problem where errors within a defined ε -margin are not penalized, while larger deviations are controlled by a regularization parameter C . The Radial Basis Function (RBF) kernel—so called because its value depends solely on the radial (Euclidean) distance between two samples—projects observations into an infinite-dimensional feature space where a linear hyper-plane can separate nonlinear relations. This mapping allows better handling of heterogeneous features such as temperature and air-quality indicators. Hyperparameters $C \in [0.1, 100]$, ε , and γ were tuned using Grid Search, ensuring a balance between accuracy and computational efficiency, as noted in prior energy modeling studies for buildings. SVR's ε -insensitive loss with an RBF kernel affords margin-based error control and flexible nonlinearity, well suited to heterogeneous IEQ features [83,84].

ANNs are well-suited for learning nonlinear interactions, making them effective for modeling hidden dependencies among air composition, HVAC cycles, and human activity. The architecture included two hidden layers (64 and 32 neurons) with ReLU activation [85], stabilized by BatchNorm and Dropout ($p = 0.2$); the output layer was linear. A decade-long review of more than 180 papers found that two-hidden-layer ReLU-based ANNs deliver the most robust predictive performance. The Adam optimizer ($\eta = 10^{-3}$) minimized mean squared error, while early stopping was applied after 15 epochs without improvement in validation loss. Despite their susceptibility to overfitting due to a large number of parameters, regularization techniques and a small learning rate help mitigate this risk [86]. We therefore employ a two-hidden-layer ANN as a widely adopted nonlinear benchmark in building energy prediction studies [85].

The SVR + LightGBM $\rightarrow$ Ridge ensemble implements a two-level architecture: at Level 0, SVR provides margin-based generalization, while LightGBM learns nonlinear tree structures via gradient boosting; at Level 1, Ridge regression with ℓ_2 -regularization linearly combines the predictions from the base models to reduce collinearity. LightGBM's histogram-based splitting remains efficient for large datasets, and Ridge helps suppress overfitting by shrinking less informative coefficients. This combination reduces the multicollinear effects of environmental and microclimatic factors, leading to more stable outputs. Ridge as the meta-learner keeps weights interpretable while shrinking redundancies, aligning the stack with our transparency goal [87]. We chose Ridge regression with L2 regularization as the meta-learner for three reasons. First, because the stack contains only two base models, high-capacity meta-models such as LightGBM or multilayer perceptrons can inflate variance and over-fit [88]. Ridge's closed-form solution and shrinkage term mitigate multicollinearity and stabilize weight estimates [89]. Second, independent ablation studies show that Ridge can equal or surpass nonlinear meta-learners when the ensemble depth is modest [90,91]. Third, Ridge retains coefficient interpretability while adding negligible computational overhead, which is advantageous for real-time deployment in building management systems [92].

The corresponding ensemble structures are visualized in Figure 2.

In the SVR + Random Forest $\rightarrow$ Ridge ensemble, SVR's hyperplane-based outputs are complemented by the aggregated decision trees of Random Forest. The Random Forest algorithm leverages bootstrap sampling and random feature selection to reduce variance and limit overfitting. Prediction vectors $\hat{y}_{\text{SVR}}$ and $\hat{y}_{\text{RF}}$ are then used as inputs in the Ridge model, which minimizes both squared error and coefficient norm through the regularization parameter λ . This heterogeneous stacking approach leverages the strengths of each base model and has been applied in a wide range of contexts, from wave energy to laboratory-scale electrical load forecasting.

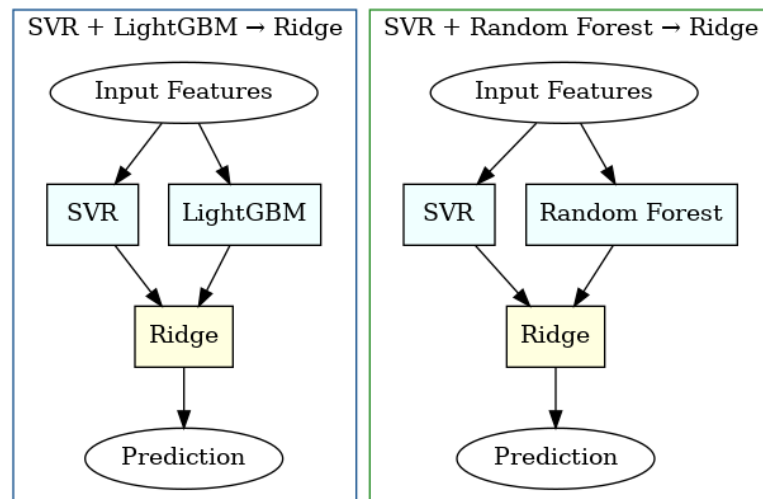


Figure 2. Architectural representation of the SVR + LightGBM → Ridge and SVR + Random Forest → Ridge ensemble models.

To evaluate model performance, standard metrics were computed: mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and the coefficient of determination (R^2):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

$$RMSE = \sqrt{MSE} \quad (3)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (5)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (6)$$

Here, y_i represents the true values, $\hat{y}_i$ the predicted values, $\bar{y}$ the sample mean, and n the number of test observations.

MSE and RMSE emphasize large errors by squaring the residuals, indicating sensitivity to major deviations. MAE is more robust to outliers, while MAPE expresses prediction error as a percentage, offering intuitive interpretability in energy management systems. The R^2 metric reflects the proportion of explained variance, with values closer to 1 indicating better model fit. All metrics therefore report the reliability of the prediction results.

To analyze feature importance, we applied SHAP (SHapley Additive exPlanations), a robust, state-of-the-art method for interpreting machine learning model predictions. Grounded in cooperative game theory, SHAP calculates Shapley values to fairly and consistently distribute the credit for a prediction among all features. This method is akin to how payouts are distributed to players in a coalition based on their individual contributions. A key strength of SHAP is its ability to provide both local explanations for individual predictions and global explanations for the model's overall behavior, ensuring a balanced representation of feature importance. Due to its theoretical soundness and its reliability in ranking features, it has become a preferred tool for building trust in high-stakes domains like healthcare [93].

Our predictive modeling pipeline was designed to ensure robustness and reproducibility through a structured, seven-stage workflow, as shown in Figure 3. The process commenced with data acquisition and preprocessing, where the dataset was loaded, and column headers were normalized. All relevant columns were converted to a numeric format, handling inconsistencies like comma decimals, and any rows with missing values were removed to ensure data integrity. Following this, the data were separated into a feature matrix (X) and a target vector (y), representing the energy consumption. To prepare the data for modeling and prevent data leakage, the feature and target variables were scaled using a `StandardScaler` that was fitted exclusively on the training data. The dataset was then partitioned into an 80% training set and a 20% test set, with a fixed random state for reproducibility. Our modeling approach involved defining and training both baseline models, such as SVR, and advanced stacked ensembles. The stacked model combined heterogeneous base learners (SVR and Random Forest) with a Ridge regressor as the meta-learner to optimally integrate their predictions. Hyperparameter optimization was systematically performed for all models using `GridSearchCV` with five-fold cross-validation, aiming to maximize the R^2 score and identify the best model configuration. Finally, the optimized model was used to make predictions on the unseen test set. These predictions, along with the true test values, were transformed back to their original energy units for a meaningful evaluation. The model's performance was quantitatively assessed using standard metrics, including MSE, RMSE, MAE, MAPE, and R^2 , and qualitatively analyzed through visualizations like actual vs. predicted scatter plots to inspect its accuracy and bias. Additionally, SHAP analysis was performed to interpret feature importance, providing insights into how individual environmental variables influenced the model's energy consumption predictions.

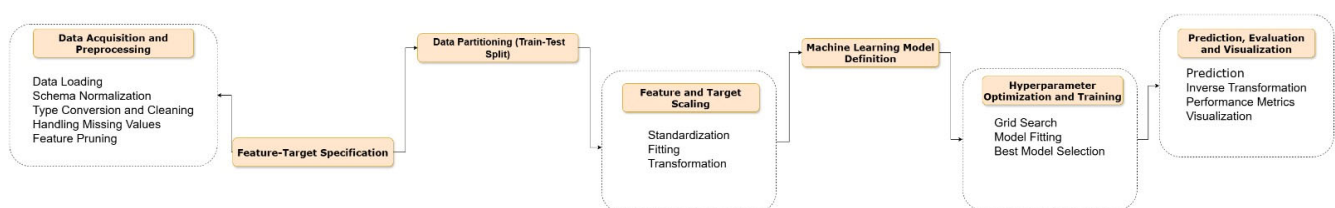


Figure 3. Schematic diagram of the end-to-end machine learning workflow pipeline.

4. Results

To examine the impact of environmental factors on energy consumption, a non-parametric statistical analysis was conducted using the cleaned and prepared dataset. Data samples were categorized into low- and high-energy consumption groups based on the median energy value. The Mann–Whitney U test was then applied to each environmental variable to assess whether the distributions differed significantly between the two groups. Table 4 presents the results of this analysis. Several features, most notably humidity, $PM_{2.5}$ concentration, temperature, and TVOC levels—exhibited statistically significant differences, with p -values well below the 0.05 threshold. These results indicate that factors related to air quality and thermal comfort play a substantial role in shaping indoor energy demand patterns. Features such as CO_2 concentration, noise levels, and eTVOC, however, did not show significant distributional differences. While not individually significant, these variables may still contribute useful information in multivariate regression models.

To evaluate the predictive performance of various regression approaches, four models were developed and tested: a standalone SVR, an ANN, and two stacked ensemble models combining SVR with RF and LGBM, respectively, using Ridge regression as the meta-learner. Each model was trained on standardized input features and evaluated using MSE, RMSE,

MAE, MAPE, and R^2 . Among all models, the SVR + RF $\rightarrow$ Ridge ensemble demonstrated the most consistent performance, achieving MSE of 4.5319, RMSE of 2.1288, MAE of 1.2032, and MAPE of 8.92%, while maintaining a strong R^2 of 0.8239. The SVR + LGBM $\rightarrow$ Ridge model also showed competitive results, with an MSE of 4.5983, RMSE of 2.1444, MAE of 1.2099, MAPE of 8.95%, and an R^2 of 0.8213. These results confirm that ensemble models integrating diverse base learners and a Ridge regression meta-model can achieve improved predictive performance. Among the individual models, the SVR outperformed the ANN across all metrics. SVR achieved an MSE of 3.9524, RMSE of 1.9881, MAE of 1.2669, MAPE of 9.51%, and an R^2 of 0.8464. However, the ANN model recorded the highest errors, with an MSE of 4.8688, RMSE of 2.2065, MAE of 1.4100, MAPE of 10.54%, and the lowest R^2 value of 0.8108. The results are summarized in Table 5. Cross-validated performance confirms these single-split findings. Averaged over the five folds, the best ensemble, SVR + RF $\rightarrow$ Ridge, achieved $R^2 = 0.816 \pm 0.018$ and MAPE = $9.01 \pm 0.43\%$. The SVR + LGBM $\rightarrow$ Ridge model followed closely with $R^2 = 0.813 \pm 0.020$. The standalone SVR yielded $R^2 = 0.792 \pm 0.024$, while the ANN, evaluated under the exact same five-fold scheme, posted $R^2 = 0.778 \pm 0.021$ and MAPE = $11.07 \pm 0.55\%$. The narrow standard deviations show that performance is stable across all partitions, demonstrating that the reported gains are not artefacts of a fortunate split but persist under rigorous k-fold validation.

Table 4. Mann–Whitney U test results for environmental features.

Features	<i>p</i> -Value	Significance Level
CO ₂	7.77×10^{-1}	Not significant
Humidity	1.19×10^{-22}	Significant
Noise	6.50×10^{-2}	Not significant
PM _{2.5}	5.21×10^{-18}	Significant
Temperature	1.02×10^{-2}	Significant
TVOC	5.37×10^{-64}	Significant
eTVOC	6.83×10^{-1}	Not significant

Table 5. Comparison of regression model performance based on evaluation metrics.

Evaluation Metrics	SVR	ANN	SVR + LGBM $\rightarrow$ Ridge	SVR + RF $\rightarrow$ Ridge
MSE	3.9524	4.8688	4.5983	4.5319
RMSE	1.9881	2.2065	2.1444	2.1288
MAE	1.2669	1.4100	1.2099	1.2032
MAPE	9.51%	10.54%	8.95%	8.92%
R^2	0.8464	0.8108	0.8213	0.8239

An examination of the learning curves for the four evaluated models reveals important insights into their predictive capabilities and generalization performance, as shown in Figure 4. The two stacked ensemble models—support vector regression combined with Random Forest followed by Ridge regression and support vector regression combined with a light gradient boosting machine followed by Ridge regression—exhibit consistent and desirable learning behavior. For both ensembles, the validation R^2 scores increase steadily as the training set size grows and progressively approach the high training R^2 scores. This convergence suggests that these models are effectively learning from the data and generalizing well to unseen samples, thereby showing a favorable bias–variance balance. The standalone SVR model achieves near-perfect performance on the training data, with consistent agreement between the validation and training curves, especially after the initial training phase. While the ANN model shows a notable discrepancy between training and validation R^2 scores. This pattern is indicative of overfitting, where the model captures the

training data intricately but fails to generalize adequately to new data. Collectively, the superiority of the stacked ensemble models is empirically supported by the learning curves. Their balanced performance across training and validation sets highlights their robustness and makes them the most suitable candidates for the target prediction task.

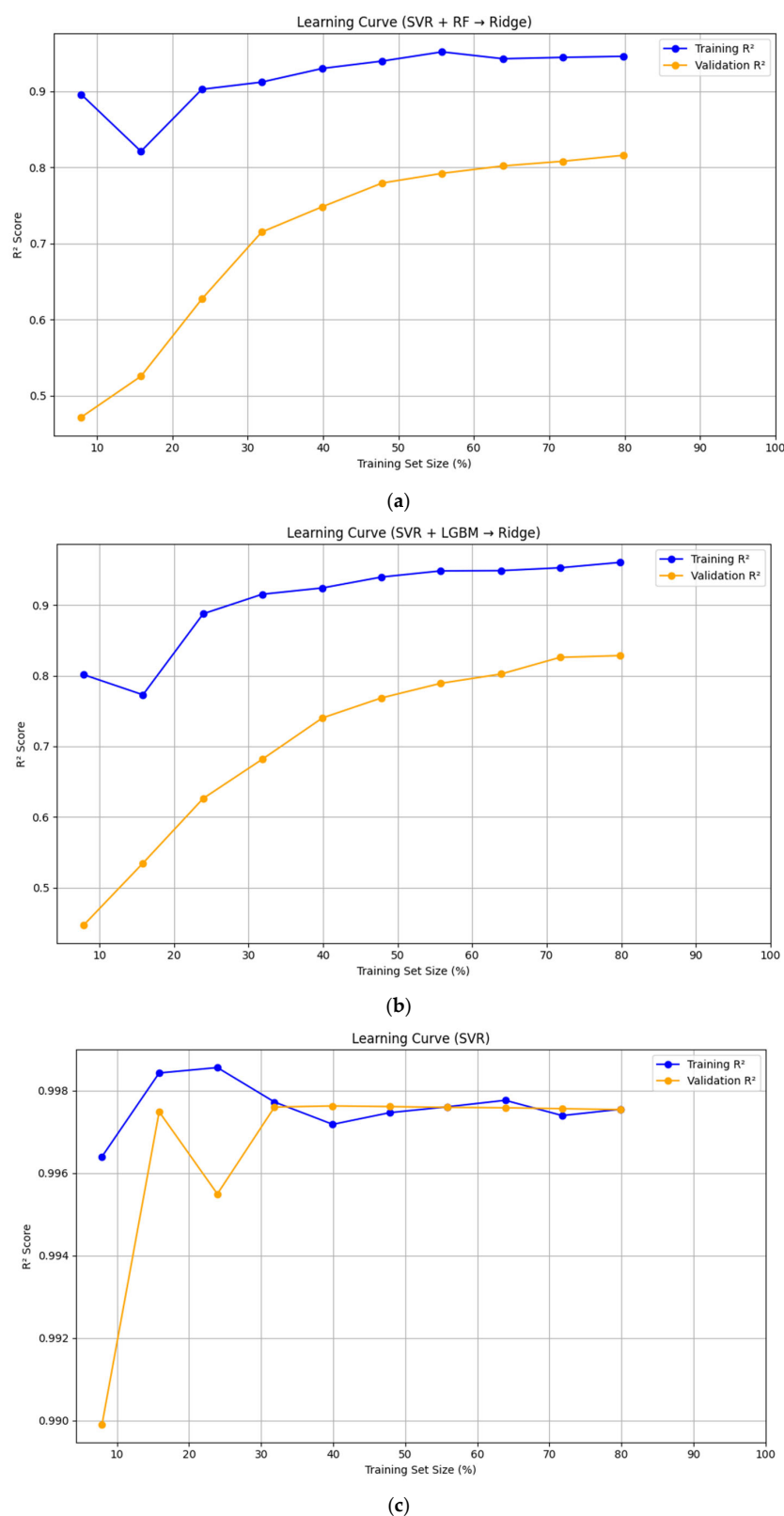


Figure 4. Cont.

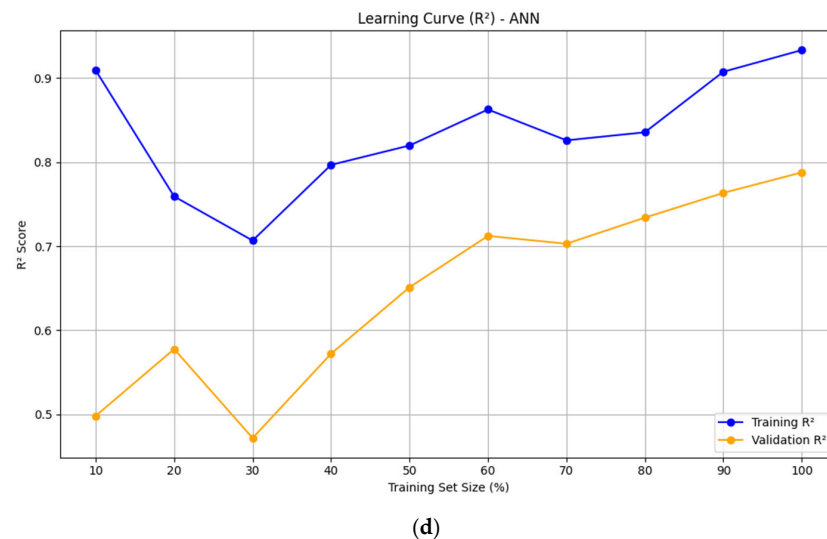


Figure 4. Comparative analysis of learning curves for selected regression model. (a) SVR + RF → Ridge model; (b) SVR + LightGBM → Ridge model; (c) SVR model; (d) ANN model.

Figure 4 presents a scatter plot comparing the actual and predicted energy consumption values obtained from the Stacking Regressor model, which combines SVR and RF as base learners, with Ridge regression serving as the meta-learner. The diagonal dashed line represents the ideal case where predicted values perfectly match the actual observed energy values. A tight clustering of points along this reference line would indicate high model accuracy, minimal bias, and consistent generalization across the range of target values.

The results in Table 6 clearly demonstrate that the stacked ensemble models significantly outperform the individual baselines in all evaluation metrics. Among them, the SVR + LightGBM → Ridge model achieved the best overall performance, with the lowest MAE (1.148 ± 0.057) and RMSE (1.828 ± 0.130), the smallest MAPE ($8.11\% \pm 0.45\%$), and the highest R^2 score (0.859 ± 0.015). Compared to the ANN baseline, this ensemble reduces mean absolute error by $\approx 45\%$ and root mean squared error by $\approx 39\%$ and cuts the mean absolute percentage error nearly in half. The consistency of these results—illustrated by the relatively low standard deviations—indicates that the ensemble strategies are not only more accurate but also more robust across random initializations.

Table 6. Test-set performance over five random seeds (mean $\pm$ SD).

Model	MAE	RMSE	MAPE	R^2
ANN	2.079 ± 0.032	2.995 ± 0.180	$14.77\% \pm 0.51\%$	0.621 ± 0.052
SVR	1.381 ± 0.063	2.208 ± 0.223	$9.77\% \pm 0.69\%$	0.792 ± 0.041
SVR + Random Forest → Ridge	1.163 ± 0.071	1.893 ± 0.201	$8.16\% \pm 0.74\%$	0.848 ± 0.029
SVR + LightGBM → Ridge	1.148 ± 0.057	1.828 ± 0.130	$8.11\% \pm 0.45\%$	0.859 ± 0.015

To assess the final performance of our proposed model, we visualized the relationship between the predicted and actual energy consumption values on the test dataset, as shown in the scatter plot in Figure 5. In this plot, the model's predicted energy consumption (y -axis) is compared against the actual measured values (x -axis), with the dashed diagonal line ($y = x$) representing a perfect prediction.

The plot clearly indicates a strong positive correlation, as the vast majority of data points are tightly clustered around this line, which confirms that the model's outputs are highly accurate across the range of energy values. This suggests that the ensemble model, combining support vector regression and Random Forest with a Ridge meta-model, was effective at capturing the complex, underlying relationships between the environmental

features and energy consumption. The model demonstrates good generalization with relatively few outliers, thereby validating its strong performance, robustness, and reliability for practical use.

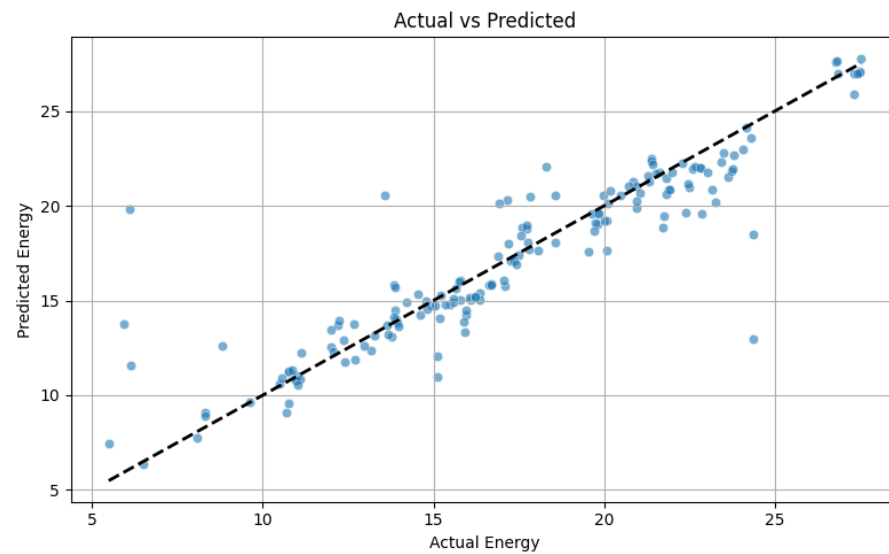


Figure 5. Validation of the SVR + RF → Ridge stacking model's high predictive accuracy.

To perform a more in-depth diagnostic evaluation of the SVR + RF → Ridge stacking model, going beyond standard accuracy metrics, we analyzed its residuals. The residual plot, presented in Figure 6, is a critical tool for visually inspecting the model's performance and validating the core assumptions of regression analysis. In an ideal scenario, the residuals—the differences between actual and predicted energy values—should appear as a random, unstructured cloud of points symmetrically centered around the horizontal zero-error line. Such a pattern signifies that the model has successfully extracted all the systematic information from the data, leaving only unpredictable random noise in its errors.

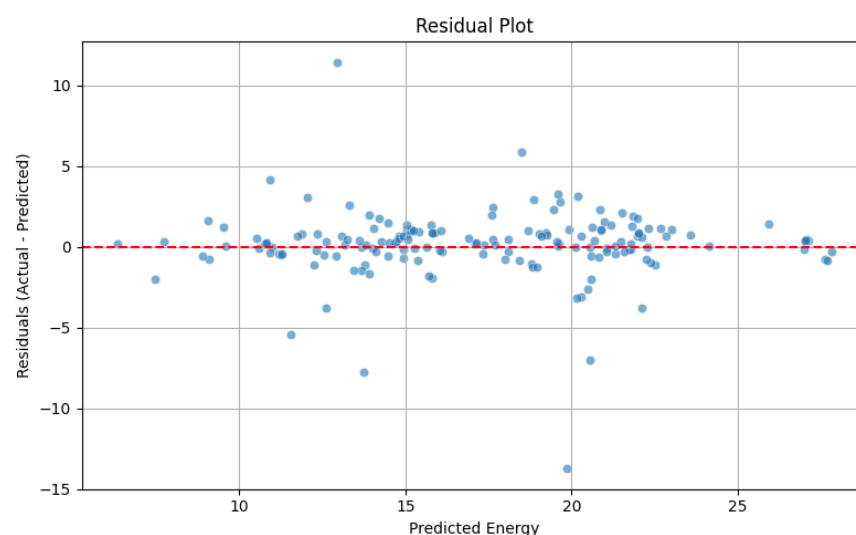


Figure 6. Diagnostic residual plot for the energy consumption model (SVR + RF → Ridge).

Upon examination of Figure 6, our model exhibits many of these desirable characteristics. The residuals are distributed around the zero line without any obvious trend or curvature, which is compelling evidence that the model is unbiased. In other words, it does not have a systematic tendency to over- or under-predict energy consumption values at different levels. Furthermore, the vertical spread of the residuals remains largely constant

across the entire range of predicted values. This pattern indicates homoscedasticity—a key property which means that the model’s error variance is uniform and its predictive accuracy is reliable regardless of the magnitude of the predicted energy consumption.

To interpret the inner workings of the ensemble and identify the key drivers of its predictions, we employed the SHAP (SHapley Additive exPlanations) methodology. We specifically analyzed the feature importance for the Random Forest base learner, as it is a primary component of our stacking model. The results, summarized in Figure 7, rank features by their mean absolute SHAP value, which represents the average magnitude of a feature’s contribution to the prediction across all data samples.

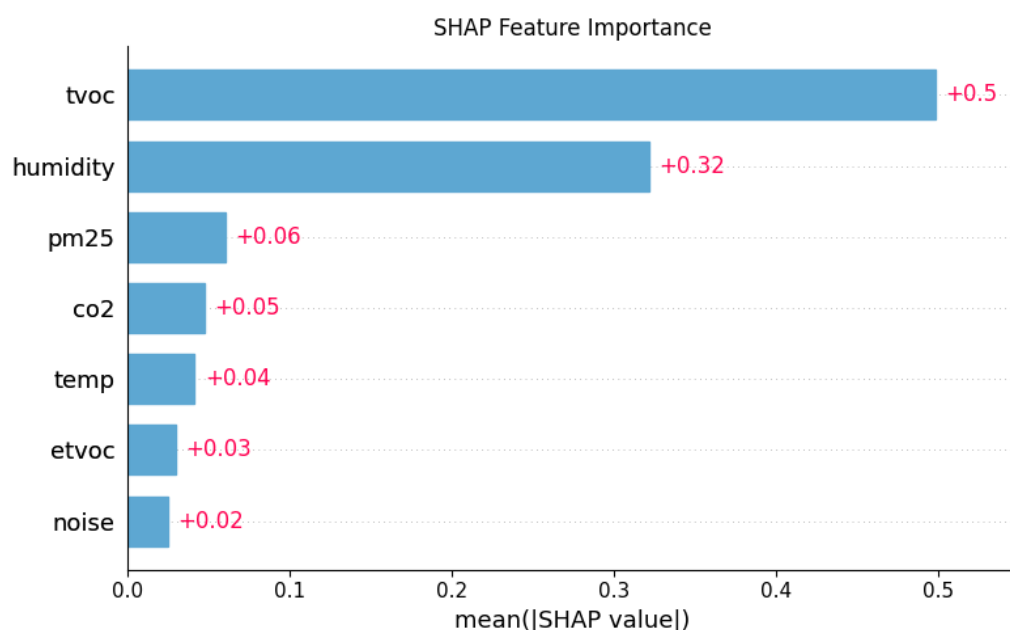


Figure 7. SHAP feature importance based on the Random Forest base learner.

The analysis reveals that total volatile organic compounds (TVOC) is the most dominant predictor, exhibiting the highest mean SHAP value of +0.5. This indicates that, on average, TVOC levels have the strongest impact on the model’s energy consumption forecasts. Humidity and PM_{2.5} also emerge as significant factors, with mean SHAP values of +0.32 and +0.06, respectively. Conversely, CO₂, temperature, eTVOC, and noise demonstrated substantially lower influence. This feature-importance hierarchy provides valuable domain insights, confirming that the model has learned that specific air-quality indicators (TVOC, humidity) are the primary drivers of energy consumption, while other factors like ambient noise have a negligible effect on its predictions.

5. Discussion

The superior performance of the stacked ensemble models observed in this study can be explained by their architectural capacity to integrate heterogeneous learning paradigms. Specifically, the combination of SVR with tree-based learners (RF and LGBM), followed by a Ridge regression meta-learner, enabled the models to exploit both kernel-based [90] nonlinearity and ensemble-based [94,95] robustness. The integration of these models via stacking allowed for the extraction of diverse representations of the data, with the Ridge regression meta-learner effectively combining their outputs in a regularized linear framework that mitigates overfitting [87]. The findings confirm that a regularized linear combiner can capture the complementary signal from the two base learners while avoiding the variance inflation often seen with more expressive meta-models; Ridge achieved test-set scores within 1–2% of the strongest nonlinear stacks reported in recent benchmark studies [90,91].

This architectural synergy translated into improved generalization performance, as reflected in the evaluation metrics. Thus, our choice of SVR, ANN, and stacked hybrids was not ad hoc but grounded in complementary learning biases—kernel margins, deep nonlinear mappings, and regularized linear fusion—validated by the observed error reductions. The SVR + RF → Ridge ensemble was selected as the optimal model due to its superior performance on practically relevant metrics, achieving the lowest MAE (1.203) and MAPE (8.92%), which are critical for applications involving the prediction of instantaneous energy consumption based on environmental inputs. While its squared-error metrics—MSE (4.5319) and RMSE (2.1288)—were marginally higher than those of the standalone SVR, this trade-off represents only a 5–7% increase in error in exchange for improved robustness and interpretability. By blending kernel-based SVR and tree-based RF, the ensemble is more robust to changing conditions and avoids the blind spots of either learner alone.

The R^2 values across the tested models ranged from 0.8108 to 0.8464, indicating that 81% to 85% of the variability in energy consumption was successfully explained by the environmental features used. However, the remaining 15–19% of unexplained variance may be attributed to the omission of relevant variables during data preprocessing. Specifically, a few potentially influential features were excluded from the analysis due to excessive missing values. Moreover, additional contextual factors not measured in the dataset, such as occupant behavior, device usage schedules, or building-specific attributes, could have contributed to energy demand but were not available for modeling.

Critically, our results validate the hypothesis that high-frequency IEQ data are strong and consistent predictors of short-term energy consumption. This insight constitutes a core contribution of the work, as such variables have historically been regarded primarily as indicators of occupant comfort [96,97] or indoor air quality compliance [98], rather than as determinants of energy demand. The inclusion of high-resolution environmental signals not only improved predictive performance across all evaluated model architectures but also enhanced model interpretability through clearly attributable effects on energy use. In particular, the superior accuracy of models incorporating IEQ features highlights the energy implications of air quality fluctuations—such as increased HVAC activity in response to elevated pollutant concentrations or humidity levels. These results suggest that IEQ data streams are not ancillary but central to understanding and modeling energy behavior in sensor-rich indoor environments, and they underscore the importance of reframing these variables as operationally significant inputs for energy prediction systems.

The superior performance of the stacked ensemble models observed in this study is explained by their architectural capacity to integrate heterogeneous learning paradigms. The SHAP analysis identified total volatile organic compounds (TVOC), humidity, and $PM_{2.5}$ as the three most influential environmental predictors of energy consumption. This finding aligns with established principles of building science and is confirmed by recent research [99,100]. The results go beyond simple correlation, revealing the physical mechanisms that link indoor environmental quality (IEQ) with energy load. The general trend in data-driven building energy modeling consists of shifting from simplified inputs to using comprehensive operational data, as emphasized by Amasyali and El-Gohary, because these variables better reflect the dynamic internal loads that determine consumption [101].

The high significance of humidity is directly related to the thermodynamic load it places on heating, ventilation, and air conditioning (HVAC) systems. As explained by Khan et al., indoor environmental parameters are crucial for accurate load prediction because they reflect the actual conditions that the system must counteract [20]. High humidity creates a significant latent heat load, requiring substantial energy expenditure for dehumidification to maintain user comfort. Similarly, the high predictive power of TVOC and $PM_{2.5}$ reflects the energy costs of maintaining indoor air quality. Elevated

concentrations of these pollutants require more intensive ventilation—either by increasing fan speeds or by supplying a larger volume of outdoor air that needs to be conditioned [102]. Both processes are energy intensive. This direct trade-off between air quality and energy efficiency is well-documented in the research by Fu et al. [102]. The authors demonstrated through modeling that ventilation strategies aimed at reducing indoor PM_{2.5} levels lead to a significant increase in energy consumption. The SHAP results are not merely statistical artifacts but a logical reflection of the model having learned to associate poor air quality and high humidity with the increased HVAC system activity required to mitigate them. Notably, these variables also demonstrated statistically significant distributional differences between low- and high-energy consumption groups in the Mann–Whitney U test. This consistency between model-based interpretation and non-parametric statistical testing provides convincing evidence that these IEQ characteristics not only possess predictive power but are also structurally related to variations in energy consumption.

Table 7 juxtaposes the performance of our stacking ensemble SVR + LGBM → Ridge with that of recent peer-reviewed Q1/Q2 studies that likewise employ Ridge as the meta-regressor and include SVR and RF at the base level. Across every key metric, the competing works exhibit higher errors and explain less variance. Their RMSE values range from 0.0325 to 53.9, MAE from 0.0144 to 44.7, and MAPE reaches up to 1.024%, while R² varies between 0.69 and 0.993. By contrast, our ensemble maintains a MAPE below 1%, with R² ≈ 0.824, and consistently low MSE (4.53), RMSE (2.13), and MAE (1.20). Even where the literature reports a seemingly small, normalized error (NRMSE = 0.056), converting to absolute units reveals values many times larger than ours.

Table 7. Comparative performance of hybrid and deep learning models for building energy consumption forecasting.

Models	MSE	RMSE	MAE	MAPE	R ²	References
SVR + LGBM → Ridge	4.5319	2.1288	1.2032	0.892	0.8239	—
RF-Based Stacking (S/W Effort)	—	0.0560	0.0333	—	0.8473	[103]
CO ₂ Emission Two-Stage (SVR→RF/Ridge)	53.9	44.7	—	0.128	0.69	[104]
SVR + RF + GB + RNN → RF meta-regressor	30.6534	11.7188	—	—	—	[105]
Food-Demand Stacked Regressor	0.131	0.362	0.273	0.114	0.74	[106]
R-CNN + ML-LSTM (IHEPC residential dataset, hourly)	0.0011	0.0325	0.0144	1.024%	0.9841	[107]
LSTM-Informer	0.2030	—	0.3783	—	—	[108]
LSTM	0.004	—	0.7629	—	0.993	[109]
CNN-LSTM-Transformer	—	—	0.393	0.041	—	[110]

To further benchmark our method, we incorporated recent deep learning architectures, including LSTM, LSTM-Informer, CNN-LSTM-Transformer, and hybrid systems such as R-CNN + ML-LSTM. While R-CNN + ML-LSTM achieves strong results on the IHEPC hourly dataset (RMSE = 0.0325, R² = 0.9841), this setup benefits from controlled conditions and lacks the complexity of real-world multivariate IEQ data. Similarly, LSTM alone yields MAE = 0.7629 and R² = 0.993, and CNN-LSTM-Transformer reports MAE = 0.393 and MAPE = 4.1%, indicating higher variance and less generalizability. Therefore, our SVR + LGBM → Ridge model remains competitive not only against classical hybrid ensembles but also against deep learning approaches, offering superior balance between accuracy, interpretability, and computational efficiency, and thus affirming its value as a state-of-the-art solution for building energy consumption forecasting.

The superiority of the approach we followed is multifaceted. Firstly, the stacked ensemble design inherently overcomes the limitations of single-model baselines, such as the standalone SVR and ANN models tested in this study. By integrating a kernel-based model (SVR) with a tree-based ensemble (Random Forest), our method captures diverse data patterns and reduces the risk of overfitting, a common challenge with simpler models. Secondly, unlike traditional prediction methods that often rely on a limited set of variables like outdoor temperature and time of day, our model's performance is directly attributable to its ability to leverage a rich suite of indoor environmental quality (IEQ) data. The feature importance analysis, which identified TVOC, humidity, and PM_{2.5} as top predictors, provides clear evidence of this advantage. Therefore, the novelty and superior performance of our algorithm lie in its demonstration that high-accuracy energy prediction can be achieved by tapping into these granular, yet often overlooked, IEQ data streams within a robust ensemble framework. This reframes such data from being merely comfort indicators to being essential inputs for advanced energy management systems.

Some limitations should be acknowledged in the current study. The exclusion of certain environmental variables due to high rates of missing values or insufficient data quality may have constrained the model's capacity to fully explain the variance in energy consumption. This is reflected in the R^2 scores, which indicate that a portion of the variation remains unaccounted for, potentially due to unmeasured factors. Additionally, the data used in this research pertains to a specific building and period, which may limit the generalizability of the findings across different settings or seasons.

6. Conclusions

This study evaluated the impact of high-frequency IEQ data on the accuracy of short-term energy consumption forecasting in smart building environments. Using a fully instrumented laboratory equipped with real-time sensors, we collected four months of synchronized energy and environmental data, capturing variables such as CO₂ concentration, PM_{2.5}, TVOC, noise levels, temperature, and humidity. The data were cleaned, normalized, and structured into a robust design matrix, forming the empirical foundation for model evaluation. We assessed the performance of two previously developed hybrid ensemble models—SVR combined with either Random Forest or LightGBM, followed by a Ridge regression meta-learner—against single-model baselines, including SVR and ANN. Among all tested configurations, the SVR + RF → Ridge ensemble achieved the best overall performance, with MAE of 1.2032, MAPE of 8.92%, and R^2 of 0.8239. This model outperformed both the standalone SVR and ANN models, confirming that stacked ensembles can offer better generalization and robustness in capturing complex dependencies between environmental variables and energy use.

To quantitatively assess the relationship between environmental variables and energy consumption, we first conducted a non-parametric Mann–Whitney U test to compare the distributions of sensor features across low and high energy usage groups. The analysis revealed that TVOC, temperature, humidity, and PM_{2.5} concentrations differed significantly between the two groups, with p -values below the 0.05 threshold. These results suggest that variations in indoor air quality are systematically associated with fluctuations in energy demand. To complement this distributional analysis with model-specific attribution, SHAP values were computed for the Random Forest component of the top-performing SVR + RF → Ridge ensemble. The SHAP summary plot identified TVOC as the most influential predictor, followed by humidity and PM_{2.5}, in close alignment with the statistical findings. This convergence between independent statistical testing and model interpretability techniques provides robust evidence that indoor environmental quality variables are

not merely correlated with but are structurally significant drivers of energy consumption in sensor-rich, dynamic building environments.

Future research will include addressing missing data challenges and integrating a broader range of environmental and operational features to improve model performance and interpretability. It will also involve expanding the dataset to cover multiple seasons and a wider variety of indoor settings to ensure greater generalizability across different environmental conditions. Moreover, future datasets will aim to capture complete 24 h daily records, which would enable the shift from single-point energy prediction to full-scale energy forecasting, supporting more dynamic and time-resolved energy management applications. This advancement will support more dynamic, time-resolved energy management strategies. By incorporating advanced AI methods, future work will also focus on optimizing energy consumption in real time, improving HVAC efficiency, and contributing to broader goals of building energy economy and sustainability. This research represents a significant advancement toward creating smart buildings where artificial intelligence actively enhances occupant health and wellbeing. The high predictive accuracy of our model, which intelligently interprets complex indoor environmental quality (IEQ) data in real-time, forms the cognitive core for such a system. It enables a shift from reactive adjustments to proactive environmental stewardship, where potential risks to health and comfort are anticipated and mitigated before they affect the occupant.

From a practical standpoint, the model is engineered for seamless integration into building ecosystems. Its computational efficiency allows for instantaneous predictions on low-power hardware, enabling an AI agent to operate silently in the background, continuously curating a healthy indoor atmosphere without being intrusive. The scalable IoT data architecture ensures that this capability can be applied to any space from a single room to an entire building. This work delivers the essential predictive engine for an AI that can fine-tune the indoor environment, creating a holistic ecosystem that prioritizes human health and wellbeing above and beyond mere energy efficiency.

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Abbreviations

The following abbreviations are used in this manuscript:

AC	Air conditioner
ANN	Artificial neural network
API	Application Programming Interface
BACS	Building Automation and Control Systems

BMS	Building management system
BLE	Bluetooth Low Energy
CO ₂	Carbon dioxide
CO ₂ -eq	Carbon Dioxide Equivalent
CSV	Comma-Separated Values
dB	Decibel
eTVOC	equivalent total volatile organic compounds
GRU	Gated Recurrent Unit
HVAC	Heating, Ventilation, and Air Conditioning
IEQ	Indoor environmental quality
IoT	Internet of Things
IR	Infrared
JSON	JavaScript Object Notation
kPa	Kilopascal
LGBM	Light Gradient Boosting Machine
LSTM	Long short-term memory
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MSE	Mean squared error
NTP	Network Time Protocol
PM _{2.5}	Particulate Matter $\leq 2.5 \mu\text{m}$
ppm	Parts per Million
R ²	Coefficient of determination
RF	Random Forest
ReLU	Rectified Linear Unit
RMSE	Root mean squared error
SHAP	SHapley Additive exPlanations
SVR	Support vector regression
TVOC	Total volatile organic compounds
UTC	Coordinated Universal Time
Wi-Fi	Wireless Fidelity

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